



Guesstimate

Asked by : Swiggy



Problem Statement : Estimate the Number of Orders delivered by Swiggy per hour in India.

Clarifying Questions :

Q1 : Do we have to calculate # of orders specific to any place ?

Ans : No consider complete India.

Q2 : Do we have to consider orders on Food Delivery business, Instamart or Genie ?

Ans : Consider it for the food delivery business.

Q3 : Do we have to consider orders for a specific user types ?

Ans : No

Q4 : Do we need to consider any specific price range of orders ?

Ans : No

Q5 : Do we have to calculate for orders from a specific food chain/restaurant ?

Ans : No

Q6 : Do we have to calculate for # of orders on a weekday or a weekend ?

Ans : Consider it to be a weekend.

Q7 : Do we have to calculate for orders delivered during peak hours or any other hours ?

Ans : Calculate for peak hours

Approach :

I am following the Top down approach.

Equation : # of orders delivered by Swiggy per hour = Total # of people ordering through Swiggy * (Avg # of orders per month)/30 * % of orders in peak hours

Total Population of India = 140 crores

Assumption : Assuming Swiggy operates only in urban cities.

Lets divide the population into Urban and Rural.

Assumption : Assuming Urban : Rural split as 30:70

of people living in Urban = 30% *140 crores = **42 crores**

We have also have to think about how many people have mobile phones to order food.

In Urban areas, lets assume **75% of people have mobile phones with internet connectivity.**

Hence # of people with mobile phones & internet connectivity= 70%*42 crores = **29 crores** (approx)

Assumption : Mostly people in the age group of 15-50 yrs would order food rest all prefer eating food at home (Above 50 yrs) or people below 15 are dependent

on others.

% of people in this age group = **60%**

Hence, **# of people in the age group (15-50 yrs)** = $60\% * 29 \text{ crores} = 17 \text{ crores}$ (approx.).

Now we have to consider about how many of these people might prefer ordering food online.

Assumption : Assuming 50% of this age group actually order food online.

Therefore, **# of people ordering food online** = $50\% * 17 \text{ crores} = 8 \text{ crores}$ (approx.)

There are lot of companies in the food delivery business like Zomato, Dominos etc.

Assumption : Lets consider Swiggy has 35% of the market share.

Hence, **# of people ordering through Swiggy** = $35\% * 8 \text{ crores} = 2.8 \text{ crores}$.

Now, we have to calculate orders made through Swiggy by these **2.8** crores.

Categorizing the users into 3 types.

1. Power User - Orders 8 meals per month
2. Active User - Orders 4 meals per month
3. Passive User - Orders 1 meals per month

Assumption : Assuming **20%** of people are the **power users**, **45%** of people are **active users** and another **35%** of the people are **passive users**.

of Power users = $10\% * 2.8 \text{ crores} \sim 0.3 \text{ crores}$

of Active users = $45\% * 2.8 \text{ crores} = 1.2 \text{ crores}$

of Passive users = $45\% * 2.8 \text{ crores} = 1.2 \text{ crores}$

Total # of orders by Power users = $0.3 \text{ crores} * 7 = 2.1 \text{ crores orders per month}$.

Total # of orders by Active users = $1.2 \text{ crores} * 3 = 3.6 \text{ crores orders per month}$.

Total # of orders by Passive users = $1.2 \text{ crores} * 1 = 1.2 \text{ crores orders per month}$.

Therefore **total orders per month** = $2.1 + 3.6 + 1.2 = 6.9 \text{ crores order per month}$ approx.

Since we have to consider orders on weekend (8 days in a month).

Assumption : Assuming **30% of the total orders are ordered during the weekend.**

Total # of orders during weekend = $6.9 * 30\% = 2.07 \text{ crores}$.

Assumption : Assuming there is equal distribution of orders in all the days of weekend.

Hence, # of orders placed during a weekend days = $2.07/8 = 25 \text{ lakhs orders}$

Swiggy delivery is active in the hours = 9 am to 11 pm = 14 hours

Peak hours of food delivery in Swiggy = 8pm to 10 pm

Assumption : Assuming **30% of the total orders are ordered in the period of 8pm to 10pm.**

Orders in peak hours = $25 \text{ lakhs} * 25\% = 6.25 \text{ lakhs}$

Assumption : Assuming there is equal distribution of orders in both the hours.

Therefore, Swiggy delivers $6.25/2 \sim 3 \text{ lakhs orders per hour (approx)}$.

Sanity Check - Swiggy used to process around 25 lakh orders daily in 2023, considering 30% growth then it will be 32.5 lakhs orders per day. Seems like our estimation is reasonable.